

MacroEdge

Intelligence Platform

Full Project Scope, Technical Architecture & Investment Plan

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1. Executive Summary

Strategic Overview

MacroEdge is a locally-hosted, AI-powered macro trading intelligence platform designed to learn global market relationships over a six-month observation window before generating predictive trade signals. The system integrates news intelligence, technical analysis, and portfolio risk management under a unified Claude AI supervision layer. The target outcome is a systematic, data-driven framework that identifies 10–15% annual return opportunities in macro-sensitive assets — commodities, ETFs, and sector equities — with clearly defined risk guardrails at every layer.

Target Return	10–15% annually on deployed capital
Primary Asset Classes	Commodities (ETFs: GLD, SLV, USO), Sector Equities, Macro ETFs
Prediction Horizon	3–7 days (short-swing), up to 2 weeks for major macro events
Brokerage API	Alpaca Markets (paper trading for 6 months, then live)
AI Supervision	Claude (claude-sonnet-4) as model health monitor and risk narrator
Total Build Timeline	18 months: 6 months learning, 6 months paper validation, 6 months live
Estimated Platform Cost	USD \$200–\$650/month at full operation (see Section 7)
CFA Alignment	Directly covers CFA L1/L2: Economics, Equity Analysis, Risk Mgmt, Portfolio Theory

2. Project Vision & Design Philosophy

Why This Approach

Most retail algorithmic trading systems fail because they act on the same information everyone else has, at the same time. MacroEdge is designed around a fundamentally different principle: **observe today, predict tomorrow**. By decoupling the learning phase from the execution phase, the platform builds a causal map of global macro relationships — understanding, for example, how a geopolitical conflict in the Middle East historically flows through to Brent crude pricing over a 5-day window, or how a surprise CPI print reshapes gold's behaviour in the following week.

The three core design principles are:

- **Causal over correlation:** The system learns why assets move, not just that they move together.
- **Patience over speed:** 6 months of observation before the first live signal. No shortcuts.
- **Human-in-the-loop:** Claude supervises models; humans approve capital deployment.

3. Phase Plan — 18-Month Roadmap

Build → Learn → Validate → Deploy

Phase 0 · M 1–2 · Foundation Setup

Environment setup, Alpaca paper account, database schema, news API keys, Claude integration scaffolding. Install Python stack. Build the web dashboard skeleton (React + FastAPI). Define the asset universe.

Phase 1 · M 2–3 · Data Infrastructure

News scraper live (RSS + NewsAPI + Reddit/r/investing). Raw OHLCV historical data pull for 5 years via Alpaca + Yahoo Finance. SQLite database seeded. Event taxonomy defined: geopolitical, monetary policy, supply chain, natural disaster, earnings.

Phase 2 · M 3–4 · News Intelligence Module

FinBERT sentiment model deployed. Named-entity recognition links news headlines to tickers. Event classifier tags each article by macro category. Historical backfill: score 5 years of archived news to build the training corpus. Dashboard shows live news feed with sentiment scores.

Phase 3 · M 4–5 · Trade Intelligence Module

Technical feature engineering: RSI, MACD, Bollinger Bands, ATR, volume z-scores, sector relative strength. First ML model (XGBoost) trained on historical OHLCV + event features. Backtesting harness built. Initial backtest results reviewed with your 30-year finance advisor.

Phase 4 · M 5–6 · Signal Fusion Engine

Sentiment score + technical score combined into a single signal vector. Logistic regression fusion layer predicts direction (up/down) + confidence over 3/5/7-day horizon. Kelly Criterion position sizer implemented. Hard risk guardrails coded: max 2% capital per trade, max 20% sector concentration, drawdown kill-switch at -8%.

Phase 5 · M 6–7 · Claude Brain Layer

Claude integrated via API as the autonomous monitor. Daily log ingestion: model accuracy, confidence calibration, portfolio exposure. Claude produces: (1) model health report, (2) risk narrative, (3) suggested parameter changes. All recommendations logged for review.

Phase 6 · M 7–12 · Paper Trading Validation

Full paper trading on Alpaca. System generates signals; paper trades executed automatically. Weekly performance review. Minimum 6-month paper track record required before live capital. Target: Sharpe ratio > 1.0, max drawdown < 8%, win rate > 52%.

Phase 7 · M 13–18 · Live Capital Deployment

Conditional on Phase 6 passing all validation gates. Start with 20% of intended capital. Ramp to full allocation over 3 months if live results match paper. CFO sign-off required at each capital increase milestone.

4. Financial Models Stack

The Theory Behind the System

These are the financial frameworks and models that govern the system's logic. Each maps directly to CFA curriculum topics, making this project a live application of your studies.

Model / Framework	Category	Purpose in Platform	CFA Topic	Phase
Causal Event Chain Mapping	Macro	War → oil → transport → inflation chain. Maps event type to expected asset movements.	Economics: Business Cycles	Ph2
Commodity Correlation Matrix	Multi-Asset	Rolling 60-day correlation: gold, silver, oil, USD index, VIX. Updates daily.	Portfolio Mgmt: Correlation	Ph3
Discounted Cash Flow (DCF) Proxy	Equity	Rate-change impact on equity valuations. Used for sector rotation signals.	Equity: DCF Valuation	Ph3
Kelly Criterion	Position Sizing	Optimal position size = edge / odds. Prevents over-concentration in any single trade.	Portfolio: Risk Mgmt	Ph4
Sharpe / Sortino Ratio	Performance	Primary model validation metrics. Sortino penalises downside volatility only.	Performance Measurement	Ph6
Value at Risk (VaR) — 95%	Risk	Daily portfolio VaR. Claude monitors breaches and alerts immediately.	Risk Mgmt: VaR	Ph5
Maximum Drawdown (MDD)	Risk	Hard kill-switch at -8% portfolio drawdown. No overrides permitted.	Portfolio Risk	Ph4
Event-Driven Return Attribution	Analytics	Post-trade analysis: which event type generated which return. Model learning feedback loop.	Performance Attribution	Ph6
Mean Reversion (Z-Score)	Technical	Commodity z-score vs 20-day mean. Flags oversold/overbought conditions for entry timing.	Technical Analysis	Ph3
Momentum Factor (12-1 Month)	Factor	12-month return minus last month. Identifies assets with persistent directional drift.	Factor Investing	Ph3

5. ML Models Stack

The Intelligence Layer

The ML stack is intentionally layered: start simple, add complexity only when the simpler model has been validated. A common failure mode is jumping to deep learning before establishing a reliable baseline — this roadmap prevents that.

Model	Type	Input Features	Output	Validation Metric	Phase
FinBERT Sentiment Classifier	NLP / Transfer Learning	News headlines, article abstracts	Sentiment: Positive / Negative / Neutral + confidence score	F1 Score > 0.78	Ph2
Named Entity Recogniser (spaCy)	NLP	Full article text	Ticker symbols, company names, commodities extracted	Precision > 0.85	Ph2
Event Classifier (Random Forest)	Classification	Headline + NER output + source metadata	Event type: Geopolitical / Monetary / Supply Chain / Earnings / Disaster	Accuracy > 80%	Ph2
Technical Feature Model (XGBoost)	Gradient Boosting	OHLCV, RSI, MACD, Bollinger, ATR, volume z-score, sector RS	3-day / 5-day directional probability (up / down)	AUC-ROC > 0.60	Ph3
Macro Causal Model (XGBoost v2)	Gradient Boosting	Event type + sentiment score + asset class + days-since-event	Expected return range over 5-day window per asset class	MAE < 1.5%	Ph3
Signal Fusion Layer (Logistic Reg.)	Ensemble / Fusion	Technical score + sentiment score + macro causal score + VIX regime	Final signal: Direction + confidence % + suggested position size	Sharpe > 1.0 in backtest	Ph4
Regime Detector (K-Means / HMM)	Unsupervised	VIX, yield curve slope, USD index, commodity basket	Market regime: Risk-On / Risk-Off / Transitional	Regime stability > 3 days avg	Ph4
Portfolio Risk Monitor (Rule-Based + Claude)	Rules + LLM	Position weights, VaR, correlation, daily P&L; log	Risk alerts, narrative summary, rebalancing suggestions	Zero missed breach alerts	Ph5

Note on Deep Learning: LSTM / Transformer models are intentionally deferred to Phase 6+ pending sufficient labelled signal data (minimum 500 validated trade events). Premature use of deep learning on small financial datasets is a primary cause of overfitting and false confidence.

6. Technology Stack

Infrastructure & Tools

Backend & API Layer

FastAPI (Python)	REST API server. Handles all data requests, Alpaca order routing, Claude calls.
APScheduler	Cron-like scheduler. Runs news scraper, model retraining, daily reports on schedule.
Alpaca-Trade-API (Python SDK)	Paper + live trading. Market data streaming, order management.
SQLite → PostgreSQL	Start with SQLite (zero cost). Migrate to Postgres if data exceeds 2GB.

ML & Data Science Layer

Python 3.11+	Core language. Libraries: pandas, numpy, scikit-learn, xgboost, statsmodels.
HuggingFace Transformers	FinBERT model hosting. Run locally (no API cost after initial download).
spaCy (en_core_web_lg)	Named entity recognition. Ticker and commodity extraction from text.
Backtrader / Vectorbt	Backtesting engine. Simulates strategy on historical data.
Matplotlib / Plotly	Chart generation for dashboard and model performance visualisation.

News & Data Ingestion

NewsAPI.org	Aggregated news from 150,000+ sources. Required subscription (see costs).
feedparser (RSS)	Free: Reuters, AP, Bloomberg RSS, FT, WSJ. Zero cost.
Yahoo Finance (yfinance)	Free historical OHLCV data. 5 years of daily data for backtesting.
PRAW (Reddit API)	r/investing, r/wallstreetbets sentiment. Free tier sufficient.

Frontend Dashboard

React 18 + Vite	Fast local dev server. Component-based dashboard.
TailwindCSS	Utility-first styling. Clean, responsive layout.
Recharts	Interactive charts: portfolio performance, signal history, risk gauges.
WebSocket (FastAPI)	Live price feeds and real-time news pushed to dashboard.

Claude AI Brain Layer

Anthropic API (claude-sonnet-4)	Model health monitoring, risk narration, code suggestions.
Prompt Engineering Layer	Structured prompts: log analysis, risk assessment, anomaly detection.

Memory / Context Management	Daily log summaries fed into context. Rolling 7-day window.
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DevOps & Local Hosting

Docker Compose	Containerises backend, frontend, database. Single command startup.
Nginx (local reverse proxy)	Serves dashboard at localhost:3000 with clean routing.
Git (local repo)	Version control. Branch per major phase. Rollback capability.

7. Cost & Budget Breakdown

Monthly Operating Costs (USD)

All costs are in USD per month. Exchange rate fluctuation (~AUD 1.55/USD) should be factored into budget planning. Most services have free tiers adequate for development phases.

Item	Provider	Free Tier	Paid Tier	Our Requirement	Est. Monthly (USD)
News API	NewsAPI.org	100 req/day	\$449/mo (Business)	Developer \$149/mo (300 req/day)	\$149
Alpaca Markets	Alpaca	Paper trading free	Live: commission-free	Free (paper). \$0 live for US stocks	\$0
Claude API	Anthropic	None	\$3/M input tokens	~500K tokens/day = ~\$45/mo	\$45
Market Data (premium)	Polygon.io	Free (delayed)	\$29/mo (real-time)	Starter \$29/mo real-time US stocks	\$29
Reddit API	Reddit	Free tier	\$0.24/1000 API calls	Free tier sufficient	\$0
Cloud Hosting (optional)	AWS / Digital Ocean	N/A	\$20–\$60/mo	Localhost Phase 0–6. Cloud optional Phase 7	\$0–\$40
Domain + SSL (if needed)	Namecheap / Cloudflare	N/A	\$12/yr + free SSL	Optional for Phase 7	\$1
HuggingFace Hub	HuggingFace	Free (local models)	\$9/mo PRO	Free — run FinBERT locally	\$0
GitHub (private repo)	GitHub	Free	\$4/mo	Free tier sufficient	\$0
TOTAL (Development Phase)				Phases 0–5 (months 1–7)	~\$224/mo
TOTAL (Full Operation)				Phase 6–7 with cloud hosting	~\$264/mo

* NewsAPI cost is the largest variable. Alternative: build a custom RSS-only scraper using Reuters, AP, FT, and Bloomberg free feeds, reducing this cost to \$0 with a 20% reduction in news coverage. Recommended to start with RSS-only and add NewsAPI only if gaps appear.

* One-time hardware consideration: FinBERT runs locally. A machine with 16GB RAM and a mid-tier GPU (NVIDIA RTX 3060 or better) is recommended for training cycles. Inference on CPU is feasible but slower.

8. Risk Assessment

Technical, Financial & Operational
Risks

Risk	Category	Level	Mitigation
Model overfitting on small dataset	ML / Technical	HIGH	Enforce train/validation/test split. Walk-forward validation only. Minimum 3 years backtest data.
News data latency (news already priced in)	Financial	HIGH	Platform targets 3-7 day horizons only. Day-0 news signals are filtered. Edge comes from sustained pattern, not speed.
Regime change (model trained on bull market, deployed in bear)	Financial	HIGH	Regime detector (K-Means/HMM) flags market regime shifts. Models retrained on recent 90 days if regime changes.
Alpaca API downtime / order failure	Operational	MEDIUM	Circuit breaker in code: if API unreachable for >5 min, all pending orders cancelled. No orphan positions.
Claude API token cost overrun	Cost	MEDIUM	Hard daily token cap coded into supervisor layer. Claude called twice daily (morning brief + risk check), not continuously.
Capital loss beyond kill-switch	Financial	MEDIUM	-8% drawdown triggers full liquidation and human review. No automated re-entry for 48 hours.
Regulatory / compliance exposure	Legal	MEDIUM	Platform is for personal use only. Alpaca is FINRA-regulated. No third-party capital involved. Consult tax advisor for AUS reporting on US trades.
Sentiment model bias (negative news bias)	ML	MEDIUM	FinBERT is finance-domain trained but can over-index on negative headlines. Calibrate on 500 manually-labelled finance articles before deployment.
Local machine failure / data loss	Operational	LOW	Daily automated backup of SQLite database to external drive. Docker volumes mapped. Weekly git commit of model checkpoints.
Scope creep / premature complexity	Project	LOW	Strict phase gates. No Phase N+1 work begins until Phase N validation criteria are met and signed off.

9. CFA & Financial Theory Alignment

Learning While Building

This project is designed to be a live laboratory for your CFA curriculum. Every system component maps to a specific study area. Building and debugging these models will reinforce theory far more effectively than passive study alone.

Platform Component	CFA Level	Study Session / Topic	Key Concept Applied
Macro Event Chain Mapping	L1 + L2	Economics: Business Cycles, Geopolitics	Fiscal/monetary transmission, commodity supply-demand elasticity
Commodity ETF Signal (GLD, SLV, USO)	L1	Alternative Investments	Commodity markets, roll yield, contango/backwardation
Sharpe / Sortino / VaR Monitoring	L1 + L2	Portfolio Management	Risk-adjusted return, downside deviation, confidence interval
Kelly Criterion Position Sizer	L2	Portfolio Management: Risk Budgeting	Optimal f, expected value, probability-weighted outcomes
DCF Proxy for Sector Rotation	L1 + L2	Equity: Valuation	Rate sensitivity, duration of equities, discount rate impact
Correlation Matrix Dashboard	L1	Quantitative Methods + Portfolio Mgmt	Covariance, correlation, diversification benefit
Backtesting Engine	L2 + L3	Quantitative Methods: Return Attribution	Survivorship bias, look-ahead bias, transaction cost modelling
FinBERT Sentiment Analysis	L1	Financial Reporting + Equity Analysis	Qualitative vs quantitative signals, information efficiency
Regime Detector	L2	Economics: Market Cycles	Expansion/contraction phases, leading indicators
Claude Risk Narrative	L3	Portfolio Management: IPS + Risk Governance	Investment Policy Statement thinking, risk tolerance articulation

10. Question Bank — Finance Expert Review

30 Questions for Your 30-Year Advisor

These questions are designed to stress-test the platform's financial assumptions with your experienced finance advisor. They are grouped by domain. Record the answers — they will directly improve the system's design.

A. Macro Event Intelligence

- Q1.** In your experience, what is the typical lag between a major geopolitical event (e.g. military conflict) and a measurable impact on Brent crude prices — and how variable is that lag?

Answer / Notes:

- Q2.** Which macro event categories do you consider most consistently predictive of gold price movement over a 5-day window?

Answer / Notes:

- Q3.** How do you distinguish between a news event that is 'already priced in' versus one that the market is still processing? What signals tell you which it is?

Answer / Notes:

- Q4.** In a risk-off environment, what is the historical behaviour of silver versus gold? Is silver a reliable safe-haven or does it behave more like an industrial metal?

Answer / Notes:

- Q5.** What is the most common mistake retail analysts make when building event-to-price models?

Answer / Notes:

B. Signal Design & Edge

- Q1.** For a 3–7 day holding horizon, which technical indicators have you found to have genuine predictive power versus those that are mostly noise?

Answer / Notes:

- Q2.** How important is volume confirmation when interpreting price signals in commodity ETFs like GLD and USO?

Answer / Notes:

- Q3.** In your view, is momentum or mean-reversion a more reliable edge for the 5-day commodity swing trade timeframe?

Answer / Notes:

- Q4.** What is your view on using RSI as a timing signal in trending vs range-bound markets? How should the system handle regime detection?

Answer / Notes:

Q5. How do you think about the Kelly Criterion in practice — do you use full Kelly, fractional Kelly, or something else? Why?

Answer / Notes:

C. Risk Management & Portfolio Design

Q1. A -8% portfolio drawdown kill-switch is built into the system. Is this threshold appropriate, too tight, or too loose for a 10–15% annual return target?

Answer / Notes:

Q2. What is your recommended maximum position size as a percentage of portfolio for a single commodity ETF bet in a high-conviction scenario?

Answer / Notes:

Q3. How do you think about correlation risk in a portfolio that simultaneously holds GLD, SLV, and mining equities? Are these genuinely diversified?

Answer / Notes:

Q4. What is your framework for sizing down positions when market volatility (VIX) is elevated? Is there a VIX threshold above which you reduce exposure?

Answer / Notes:

Q5. How would you structure the risk guardrails differently for a \$10K portfolio versus a \$100K portfolio?

Answer / Notes:

D. Model Validation & Backtesting

Q1. What is the minimum number of backtested trades you would want to see before trusting a signal model's statistics (win rate, Sharpe, etc.)?

Answer / Notes:

Q2. How do you control for look-ahead bias in a backtest that uses news sentiment data — given that article timestamps can be unreliable?

Answer / Notes:

Q3. What Sharpe ratio would you consider the minimum acceptable threshold for moving from paper trading to live capital?

Answer / Notes:

Q4. What are the most dangerous forms of overfitting in event-driven trading models, in your experience?

Answer / Notes:

Q5. If a model performs well in backtest but poorly in paper trading, what are the first three things you would investigate?

Answer / Notes:

E. Alpaca API & Execution

- Q1.** For a locally-hosted retail system trading US equity ETFs via Alpaca, what are the primary execution risks we should design around?

Answer / Notes:

- Q2.** Slippage and bid-ask spread are typically ignored in backtests. For ETFs like GLD and USO, how material are these costs in practice?

Answer / Notes:

- Q3.** What order types would you recommend for this system — market orders, limit orders, or stop-limit? Why?

Answer / Notes:

- Q4.** Is there any regulatory concern with running an automated trading system on a personal Alpaca account from Australia?

Answer / Notes:

- Q5.** What data does Alpaca provide natively for risk monitoring, and what gaps would we need to fill from external sources?

Answer / Notes:

F. Platform Strategy & AI Brain

- Q1.** In your view, what are the most important early warning signals that a trading model is beginning to degrade and needs retraining?

Answer / Notes:

- Q2.** How would you use an AI system like Claude in the context of a trading desk — purely as an analyst tool, or would you trust it for any decision-making?

Answer / Notes:

- Q3.** What financial metrics should be in a daily portfolio health dashboard for a system targeting 10–15% annual returns?

Answer / Notes:

- Q4.** If the system generates a high-confidence signal that contradicts your own macro view, how do you think about resolving that conflict?

Answer / Notes:

- Q5.** Looking at the full architecture described, what is the single biggest risk you see that is not adequately mitigated in the current design?

11. Appendix — Key Terms Glossary

For Reference

Alpaca Markets	A commission-free US stock brokerage with REST + WebSocket API. Supports paper trading (simulated) and live trading.
AUC-ROC	Area Under the ROC Curve. Measures ML classifier performance. > 0.6 is a meaningful edge for financial prediction; > 0.7 is strong.
Backtesting	Running a trading strategy on historical data to estimate performance. Subject to look-ahead bias and survivorship bias if not carefully designed.
FinBERT	A BERT-based natural language model fine-tuned on financial text. Outputs sentiment (positive/negative/neutral) with high accuracy on financial news.
Kelly Criterion	A formula for optimal bet sizing: $f^* = (bp - q) / b$ where b = odds, p = win probability, $q = 1-p$. Maximises long-run capital growth.
Maximum Drawdown	The largest peak-to-trough decline in portfolio value over a period. A key risk metric. This platform hard-limits at -8%.
Mean Reversion	The tendency of an asset's price to return to its historical average after deviating. Works best in range-bound, low-trend markets.
Momentum	The tendency of assets that have performed well recently to continue outperforming. Works best in trending markets.
NER (Named Entity Recognition)	An NLP technique that extracts structured entities (company names, tickers, locations) from unstructured text.
Regime Detection	Classifying the current market environment (risk-on, risk-off, volatile, trending). Different strategies perform differently across regimes.
Sharpe Ratio	Return above the risk-free rate, divided by portfolio standard deviation. Higher is better. > 1.0 is the minimum target for this system.
Signal Fusion	Combining multiple independent signals (sentiment + technical + macro) into a single weighted output signal with higher reliability.
Sortino Ratio	Like Sharpe, but only penalises downside volatility. Better for asymmetric strategies. This platform tracks both.
VaR (Value at Risk)	The maximum expected loss over a given time period at a given confidence level. E.g. 95% 1-day VaR = the worst daily loss we expect 95% of the time.
Walk-Forward Validation	A backtesting method where the model is trained on data up to period T , then tested on $T+1$, $T+2$ etc. Prevents look-ahead bias.
XGBoost	Extreme Gradient Boosting. A high-performance ML algorithm well-suited to tabular financial data. The preferred baseline model for this system.

12. Document Sign-Off & Next Steps

CFO Review Checklist

This document is prepared for CFO review prior to Phase 0 project initiation. The following items require formal approval before expenditure commences:

- Monthly operating budget approved: USD \$224/month (development) → \$264/month (full operation)
- Alpaca paper trading account creation authorised
- NewsAPI.org subscription (\$149/mo) approved for a 3-month trial period
- Anthropic API key provisioned for Claude integration (~\$45/month)
- Polygon.io real-time data subscription (\$29/month) approved
- Hardware review completed: confirm 16GB+ RAM machine available for local model training
- 6-month paper trading gate confirmed: no live capital deployment without CFO re-sign-off
- Finance expert (30yr experience) consultation session scheduled using Question Bank (Section 10)
- CFA study plan integrated: platform build phases aligned with relevant exam study sessions
- Project review cadence agreed: monthly check-in on model performance and budget actuals

Prepared by

Finance Expert Review

CFO Approval

Name / Date

Name / Date

Name / Date